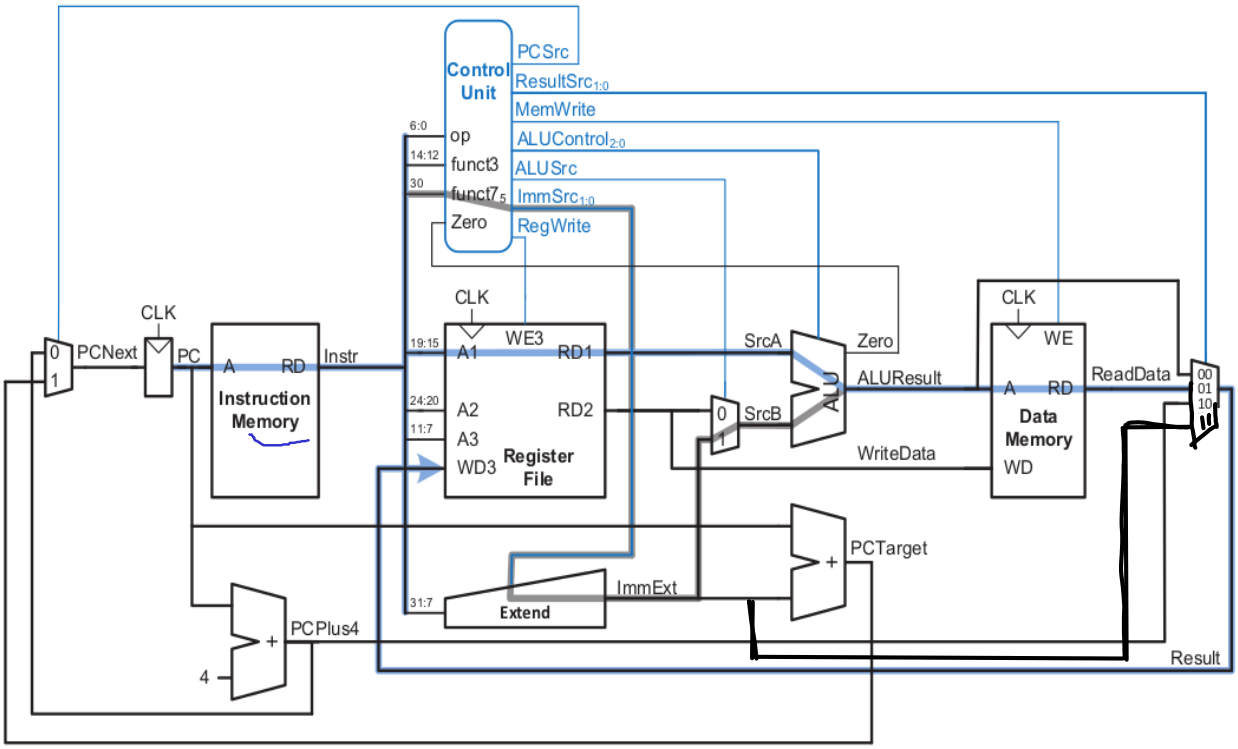
Instruction	Opcode	RegWrite	ImmSrc	ALUSrc	MemWrite	ResultSrc	Branch	ALUOp	Jump
lw	0000011	1	000	1	0	01	0	00	0
sw	0100011	0	001	1	1	xx	0	00	0
R-type	0110011	1	xx	0	0	00	0	10	0
beq	1100011	0	010	0	0	xx	1	01	0
I-type ALU	0010011	1	000	1	0	00	0	10	0
jal	1101111	1	011	x	0	10	0	xx	1
lui	0110111	1	100	x	0	11	0	xy	0
bne	1100011	0	010	0	0	xx	1	01	0



ALUOp	funct3	{op₅, funct7₅}	ALUControl	Instruction
00	x	x	000 (add)	lw, sw
01	x	x	001 (subtract)	beq
10	000	00, 01, 10	000 (add)	add
	000	11	001 (subtract)	sub
	010	x	101 (set less than)	slt
	110	x	011 (or)	or
	111	x	010 (and)	and

ImmExt = { Instr[31:12], 12'b0 }

